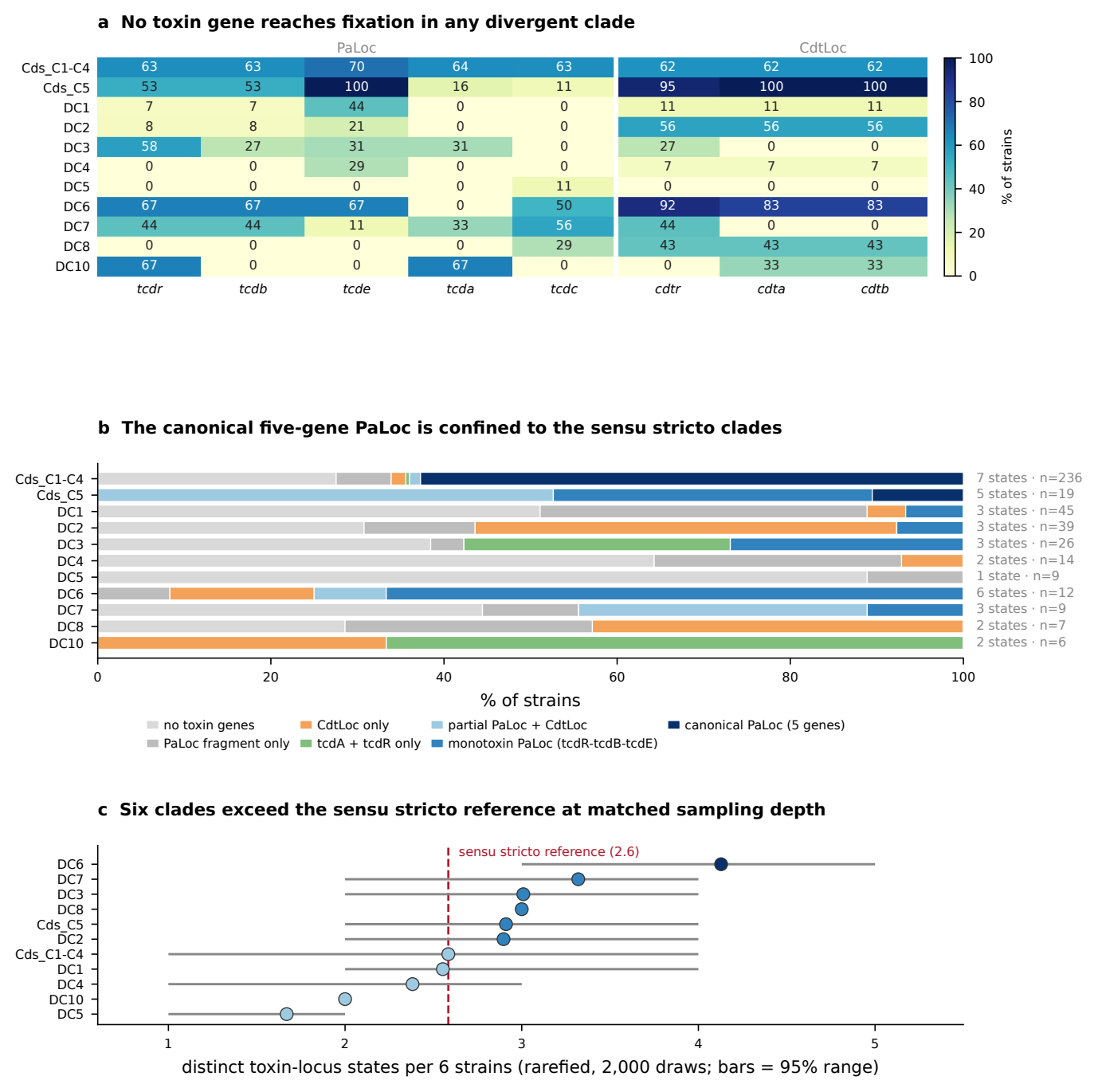
a No toxin gene reaches fixation in any divergent clade
PaLoc
CdtLoc
Cds_C1-C4 63 63 70 64 63 62 62 62
Cds_C5 53 53 100 16 11 95 100 100
DC1 7 7 44 0 0 11 11 11
DC2 8 8 21 0 0 56 56 56
DC3 58 27 31 31 0 27 0 0
DC4 0 0 29 0 0 7 7 7
DC5 0 0 0 0 11 0 0 0
DC6 67 67 67 0 50 92 83 83
DC7 44 44 11 33 56 44 0 0
DC8 0 0 0 0 29 43 43 43
DC10 67 0 0 67 0 0 33 33
tcdr tcdb tcde tcda tcdc cdtr cdta cdtb
% of strains
100 80 60 40 20 0
b The canonical five-gene PaLoc is confined to the sensu stricto clades
Cds_C1-C4 7 states · n=236
Cds_C5 5 states · n=19
DC1 3 states · n=45
DC2 3 states · n=39
DC3 3 states · n=26
DC4 2 states · n=14
DC5 1 state · n=9
DC6 6 states · n=12
DC7 3 states · n=9
DC8 2 states · n=7
DC10 2 states · n=6
0 20 40 60 80 100
% of strains
no toxin genes
CdtLoc only
partial PaLoc + CdtLoc
canonical PaLoc (5 genes)
PaLoc fragment only
tcdA + tcdR only
monotoxin PaLoc (tcdR-tcdB-tcdE)
c Six clades exceed the sensu stricto reference at matched sampling depth
sensu stricto reference (2.6)
DC6
DC7
DC3
DC8
Cds_C5
DC2
Cds_C1-C4
DC1
DC4
DC10
DC5
1 2 3 4 5
distinct toxin-locus states per 6 strains (rarefied, 2,000 draws; bars = 95% range)